

Life Cycle Assessment of a cosmetic foundation

Environmental impact analysis and sustainable redesign proposal

Author: Silvia Camagna

1. Abstract	1
2. Introduction	2
3. Methodology	2
3.1 Product Description	2
3.2 System Boundary	3
3.3 Functional Unit	3
3.4 Data Sources	4
4. Results	4
4.1 Raw Materials	4
4.2 Manufacturing	4
4.3 Packaging Production	4
4.4 Transportation	5
4.5 Use phase	6
4.6 End-of-life	6
5. Discussion	6
6. Alternative design proposal	7
7. Conclusions	8
8. References	8

1. Abstract

This study presents a simplified Life Cycle Assessment (LCA) of the Estée Lauder Double Wear Sheer Long-Wear Makeup SPF20 (30 ml), a liquid foundation previously available on the commercial market and since discontinued by the manufacturer, made in Belgium. Following a cradle-to-grave approach and the ISO 14040/14044 framework, the environmental impact of six life cycle stages was estimated using publicly available emission factor databases.

The total carbon footprint was estimated at 0.2055 kg CO₂e per functional unit. Packaging production was identified as the dominant hotspot, contributing 52.7% of total emissions, primarily due to the energy-intensive aluminium coating and multi-material packaging composition. Raw materials extraction (20.9%) and manufacturing (18.2%) represented the second and third largest contributors. Based on the findings, three ecodesign interventions were proposed: elimination of the aluminium coating, transition to mono-material packaging, and incorporation of recycled PET. Combined, these measures could reduce the product's carbon footprint by an estimated 35-40%.

2. Introduction

The global cosmetic market generates an estimated 120 billion units of packaging annually, the majority of which ends up in landfill. The cosmetic industry is one of the fastest-growing consumer sectors worldwide. Despite increasing attention to sustainability, the environmental impacts associated with cosmetic products - particularly packaging waste, resource consumption, transportation, and end-of-life disposal - remain significant challenges.

This project applies a simplified Life Cycle Assessment (LCA) methodology to a commercially available liquid foundation, the Estée Lauder Double Wear Sheer Long-Wear SPF20. The objective is to estimate the product's environmental impact across key stages of its life cycle and to identify opportunities for reducing its carbon footprint through more sustainable design choices.

The study follows a cradle-to-grave approach and relies on publicly available Life Cycle Assessment databases and environmental datasets. Based on the findings, an alternative packaging design will be proposed using ecodesign principles to improve the product's overall sustainability performance.

3. Methodology

Life Cycle Assessment (LCA) is a standardized methodology for evaluating the environmental impacts of a product across its entire life cycle, from raw material extraction to end-of-life disposal. This study follows the ISO 14040/14044 framework, which defines four phases: goal and scope definition, life cycle inventory (LCI), life cycle impact assessment (LCIA), and interpretation.

Due to data limitations and the scope of this independent study, a simplified LCA approach was adopted, focusing primarily on carbon footprint (CO₂ equivalent emissions) and packaging material impacts.

3.1 Product Description

The product selected for this LCA is the Estée Lauder Double Wear Sheer Long-Wear Makeup SPF20, a liquid foundation manufactured in Belgium and distributed across the EU market.

The product is contained in a 30 ml PET bottle with a PP cap and a metal-coated closure. The formulation includes water, silicones (dimethicone, trimethicone), titanium dioxide, iron oxides, glycerin, and several botanical extracts.

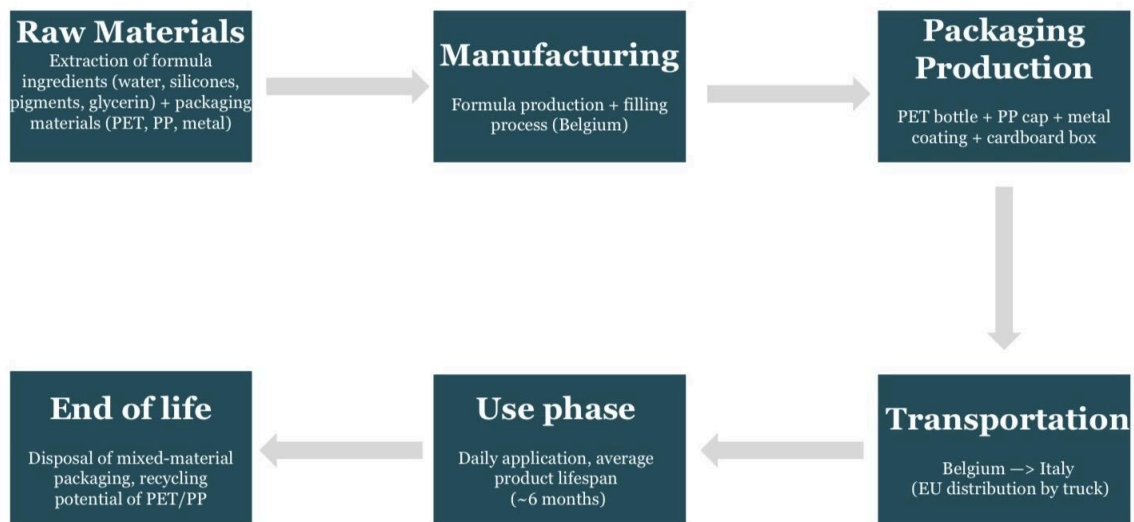
The product carries the Green Dot symbol, indicating the manufacturer's financial contribution to a European packaging recovery scheme.

3.2 System boundary

The system boundary (illustrated in Figure 1) of this LCA follows a cradle-to-grave approach, encompassing six life cycle stages:

1. Raw Materials Extraction - extraction of formula ingredients (water, silicones, pigments, glycerin) and packaging materials (PET, PP, metal).
2. Manufacturing - formula production and filling processes at the Belgian production facility.
3. Packaging production - production of PET bottle, PP cap, metal coating and cardboard box.
4. Transportation - distribution from Belgium to Italy by truck (EU road transport).
5. Use phase - daily application over an average product lifespan of approximately 6 months.
6. End-of-life - disposal of mixed-material packaging and recycling potential of PET and PP components.

Figure 1: *System boundary diagram - Life Cycle Assessment of Estée Lauder Double Wear Sheer Long-Wear SPF20 (30 ml)*



3.3 Functional unit

One unit (30 ml) of Estée Lauder Double Wear Sheer Long-Wear Makeup SPF20.

3.4 Data sources

The following publicly available databases and sources were consulted to obtain Life Cycle Inventory (LCI) data:

- Ecoinvent 3.x (via OpenLCA) - primary source for material and energy flow data,
- European Environment Agency (EEA) - transport emission factors,
- Carbon Cloud - carbon footprint estimates for cosmetic ingredients,
- Franklin Associates/PE International - packaging material LCA data,
- Peer-reviewed literature (Google Scholar) - LCA studies on cosmetic products.

Due to the complexity of the formulation, this study applies a mass-based cut-off criterion, focusing on the two ingredient categories with the most significant environmental impact: silicones and titanium dioxide, which together represent approximately 20% of the formula's mass but contribute disproportionately to its carbon footprint. Minor components (water, glycerin, botanical extracts) were excluded from detailed quantification due to their negligible individual impact and limited data availability.

4. Results

4.1 Raw materials

The raw materials extraction phase contributes an estimated 0.0429 kg CO₂e per functional unit, representing approximately 20.9% of the total carbon footprint. The two primary contributors identified within this phase are silicones (dimethicone group), accounting for 0.0369 kg CO₂e, and titanium dioxide, contributing 0.006 kg CO₂e.

4.2 Manufacturing

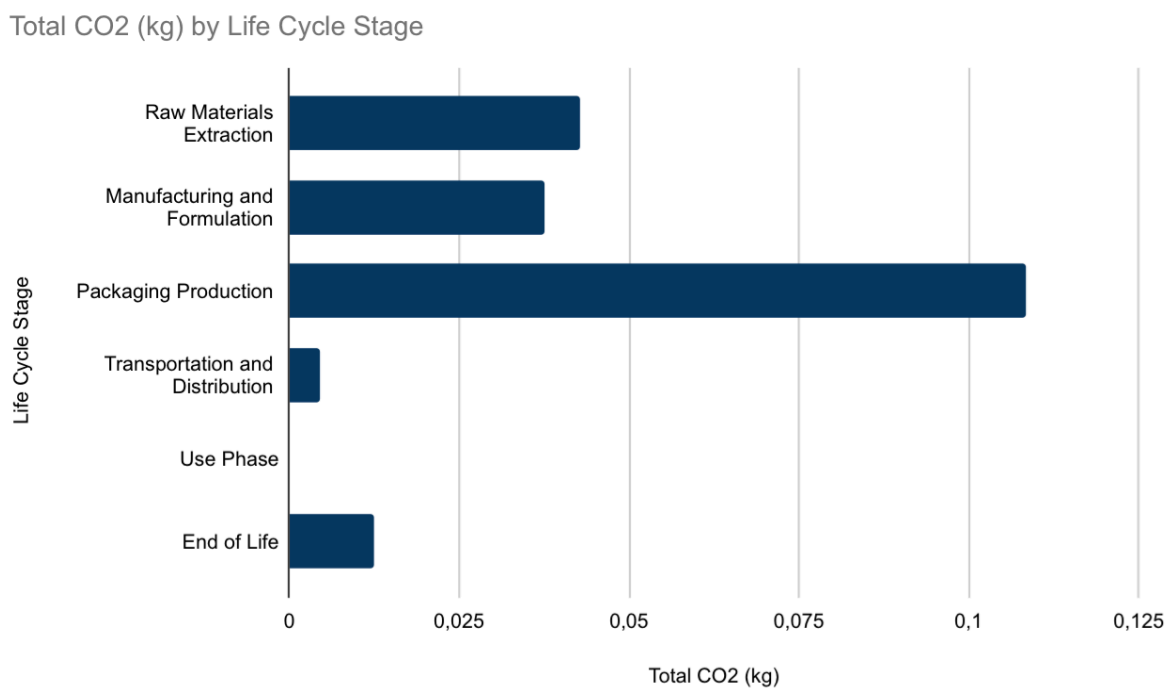
The manufacturing phase, encompassing formula mixing, heating, homogenization, and filling operations at the Belgian production facility, contributes 0.0375 kg CO₂e per unit (18.2% of the total). The estimate is based on a factory-gate emission factor of 1.25 kg CO₂e per kg of bulk liquid, derived from industry data published by the Royal Society of Chemistry.

4.3 Packaging production

With an impact of 0.1083 kg CO₂e per unit, packaging production emerges as the primary environmental burden, accounting for 52.7% of the total carbon footprint. This phase encompasses the production of four distinct components: the PET bottle (0.0487 kg CO₂e), the PP cap structure (0.0251 kg CO₂e), the aluminium metal coating (0.0210 kg CO₂e), and the cardboard outer box (0.0135 kg CO₂e).

The dominance of this phase is primarily attributable to the energy-intensive nature of primary aluminium production and the multi-material composition of the packaging, which significantly complicates end-of-life recycling.

Figure 2: Carbon footprint by life cycle stage (kg CO₂e per functional unit)



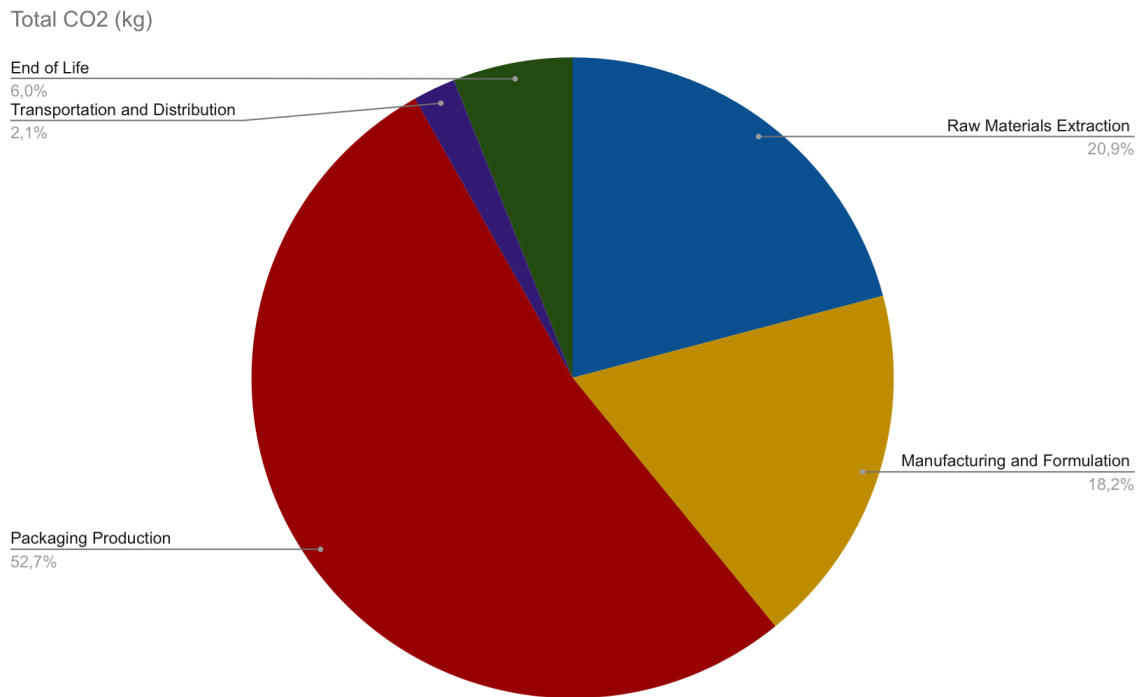
4.4 Transportation

Transportation from the Belgian manufacturing facility to the Italian distribution market contributes 0.0044 kg CO₂e per unit (2.1% of total), calculated using an average EU road freight emission factor of 90 g CO₂e per tonne-kilometre over an estimated distance of 905 km.

4.5 Use phase

The use phase generates negligible direct CO₂ emissions, as product application requires no energy input and produces no measurable gaseous emissions. This phase was therefore assigned a value of approximately 0 kg CO₂e per functional unit.

Figure 3: Carbon footprint distribution by life cycle stage (%)



4.6 End-of-life

End-of-life disposal of the packaging components contributes 0.0124 kg CO₂e per unit (6% of total), assuming mechanical recycling of both PET and PP components via the European Green Dot collection system. The aluminium coating, being inseparably fused to the PP cap structure, was excluded from the recycling scenario due to its practical non-separability, a significant design limitation identified in this study.

5. Discussion

The results of this simplified LCA indicate a total carbon footprint of approximately 0.2055 kg CO₂e per functional unit.

The breakdown of results confirms that the packaging system is the main driver of environmental impact, generating over half of the product's total emissions (52.7%). This is primarily driven by the energy-intensive production of the aluminium metal coating (7.0 kg CO₂e/kg) and the multi-material composition of the packaging system (PET + PP + aluminium), which not only increases production emissions but also significantly limits end-of-life recyclability.

Raw materials extraction and manufacturing represent the second and third largest contributors (20.9% and 18.2% respectively), driven predominantly by the high carbon intensity of silicone production (8.2 kg CO₂e/kg).

Transportation and end-of-life disposal contribute marginally to the overall footprint (2.1% and 6% respectively), suggesting that logistics optimization alone would yield limited environmental benefit.

These findings are consistent with broader LCA literature on cosmetic products, which consistently identifies packaging as the primary environmental hotspot. The multi-material nature of premium cosmetic packaging - designed primarily for aesthetic appeal and commercial differentiation rather than recyclability - represents a structural challenge for the industry's sustainability transition.

A key limitation of this study is the reliance on generic industry average emission factors rather than product specific primary data. Additionally, the mass-based cut-off criterion applied to formula ingredients introduces uncertainty in the raw materials phase. These limitations are inherent to simplified LCA methodology and should be considered when interpreting the results.

6. Alternative design proposal

Based on the hotspot analysis conducted in this study, packaging production was identified as the primary target for environmental improvement. The following redesign proposal applies ecodesign principles to reduce the carbon footprint of the Estée Lauder Double Wear Sheer Long-Wear SPF20 while maintaining its functional and aesthetic performance.

Proposal 1 - Replace aluminium coating with printed labelling

The aluminium metal coating on the PP cap represents a significant source of emissions (7.0 kg CO₂e/kg) and renders the cap non recyclable due to material inseparability. Replacing the metallic coating with high quality printed labelling or in-mould labelling would eliminate this component entirely, reducing packaging emissions by approximately 19.4% while maintaining premium visual appeal.

Proposal 2 - Transition to mono-material packaging

Building on *Proposal 1*, the current PET bottle and PP cap cannot be recycled together, transitioning to a mono-material PP packaging system would enable effective mechanical recycling at end-of-life, reducing end-of-life emissions and increasing circularity.

Proposal 3 - Incorporate recycled PET (rPET)

Replacing virgin PET with recycled PET (rPET) would reduce the bottle's carbon footprint from 3.50 kg CO₂e/kg to approximately 0.69 kg CO₂e/kg - an 80% reduction in packaging production emissions for this component alone.

Combined, these three interventions could reduce the total product carbon footprint by an estimated 35-40%, from 0.2055 kg CO₂e to approximately 0.12-0.13 kg CO₂e per functional unit, without compromising product performance or significantly altering manufacturing processes.

It is worth noting that the discontinuation of this specific formulation by Estée Lauder - as indicated in the "Gone But Not Forgotten" section on [esteelauder.com](https://esteelauder.com/discontinued-products) (<https://esteelauder.com/discontinued-products>) - may itself reflect growing market pressure toward more sustainable product lines, making the redesign proposed in this study particularly timely.

7. Conclusions

This study represents a simplified Life Cycle Assessment of the Estée Lauder Double Wear Sheer Long-Wear Makeup SPF20 (30 ml), estimating a total carbon footprint of 0.2055 kg CO₂e per functional unit.

In conclusion, packaging remains the critical life cycle stage, responsible for over half of total greenhouse gas emissions, driven by the aluminium coating and multi-material composition. Silicone production emerged as the key driver within the formulation phase.

Three ecodesign interventions were proposed: elimination of the aluminium coating, transition to mono-material packaging, and incorporation of recycled PET.

Combined, these measures could reduce the product's carbon footprint by an estimated 35-40%.

This project demonstrates that applying LCA methodology to cosmetic products can yield actionable insights for more sustainable product design - an area that remains underexplored relative to the industry's scale and environmental impact.

8. References

1. **Hedgehog LCA (2023)** - PET plastic LCA data, industry midpoint values.
2. **PlasticsEurope/Global Industry Average** - PP production carbon footprint data.
3. **European Aluminium (2022)** - Environmental profile report, primary aluminium production, EU average.
4. **Pro Carton/Carbon Footprint Study** - Folding cartons fossil carbon footprint averages.
5. **Global Silicones Council (GSC)** - Life cycle inventory data for dimethicone (PDMS) production.
6. **ScienceDirect/MDPI** - Titanium dioxide production LCA, chloride process, global average.
7. **Royal Society of Chemistry** - Carbon footprint of cosmetic manufacturing processes.
8. **European Environment Agency (EEA)** - EU road freight transport emission factors.
9. **ScienceDirect** - End-of-life emissions for PET and PP mechanical recycling.
10. **ISO 14040:2006** - Environmental management, LCA principles and framework.
11. **ISO 14044:2006** - Environmental management, LCA requirements and guidelines.